ITDS 124 Mathematics of Computing

Chapter 1. Fundamentals

1.1 Sets and Subsets

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Sets

- A set is any well-defined collection of objects.
  - Well-defined means that it is possible to decide if a given object belongs to the collection or not.
  - Objects in a set are called the elements or members of the set.

Set notation

- Listing elements of the set between braces, example $\{1,2,3\}$
- Sets are represented with uppercase letters A, B, C
- Members (or elements) of sets are represented with lowercase letters a, b, c, x, y, z, t
- Note on Brackets (Parentheses)

( )	parentheses or "round brackets“	วงเล็บ
{ }	braces or "curly brackets“	วงเล็บปีกกา
[ ]	"square brackets" or "box brackets“	วงเล็บก้ามปู
< >	"angle brackets“	วงเล็บแหลม หรือ วงเล็บสามเหลี่ยม

https://th.wikipedia.org/wiki/นขลิขิต

Membership

- Membership : $x \in A$ → x is an element of the set A
- Not a membership : $y \notin A$ → y is not an element of the set A
- EX. Let $A = \{1,3,5,7\}$. $1 \in A, 3 \in A$, but $4 \notin A$.
- EX. Let $B = \{\{1,3\},\{5,7\}\}$. $\{1,3\} \in B$, but $5 \notin B$.

Representing a set using its property

- Another way to explain set is to use the notation $\mathrm{P}(x)$.
- $\mathrm{P}(x)$ is a sentence or statement concerning the variable object x.
- The set defined by $\mathrm{P}(x)$, written $\{x \mid \mathrm{P}(x)\}$ ➔ the set of all x such that $\mathrm{P}(x)$.

- Ex. $\{x \mid x \text{ is a positive integer less than } 4\}$ is the set $\{1,2,3\}$.
- Ex. The set consisting of all letters in the word “mahidol” can be denoted by $\{m, a, h, i, d, o, l\}$ or by $\{x \mid x \text{ is a letter in the word "mahidol"}\}$.

Useful sets and their notations to keep in mind

- Positive integers: $\mathbb{Z}^+ = \{x \mid x \text{ is a positive integer}\} = \{1,2,3,\dots\}$
- Natural numbers: $\mathbb{N} = \{x \mid x \text{ is a positive integer or zero}\} = \{x \mid x \text{ is a natural number}\} = \{0,1,2,3,\dots\}$
- Integers: $\mathbb{Z} = \{x \mid x \text{ is an integer}\} = \{\dots,-3,-2,-1,0,1,2,3,\dots\}$
- Rational numbers: $\mathbb{Q} = \{x \mid x \text{ is a rational number}\}$
  - $\mathbb{Q}$ is a set with rational numbers that can be written as $\frac{a}{b}$, where a and b are integers and b is not 0, such as $-1,-2,-3,0,1,2,3,\frac{2}{3},\frac{-2}{3},\frac{2}{-5},\frac{22}{7}$
- Real numbers: $\mathbb{R} = \{x \mid x \text{ is a real number}\}$
  - $\mathbb{R}$ is a set with rational numbers and irrational numbers (i.e. $\sqrt{2},\pi$).
- Empty set: $\emptyset = \{\,\}$

Examples

- $A = \{1,3,5,7,9,11\} = \{x \mid x \text{ is } \ldots\ldots\ldots\ldots\ldots\}$
- $B = \{3,6,9,12,\ldots\} = \{x \mid x \text{ is } \ldots\ldots\ldots\ldots\ldots\}$
- $C = \{x \mid x \text{ is an integer and } 5 \leq x \leq 10\} = \{\ldots\ldots\ldots\ldots\ldots..\}$
- $D = \{x \mid x^2 - 5x + 6 = 0\} = \{\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots.\}$
- $E = \{x \mid x \text{ is a real number and } x^2 = -9\} = \{\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots..\}$
- $F = \{0,4,8,12,16,\ldots\} = \{\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\}$

Set equality

- Two sets A and B are equal if they have the same elements ➔ $A = B$
- Order of elements in those equal sets are not important.

- Ex. If $A = \{1,2,3\}$ $B = \{3,2,1\}$

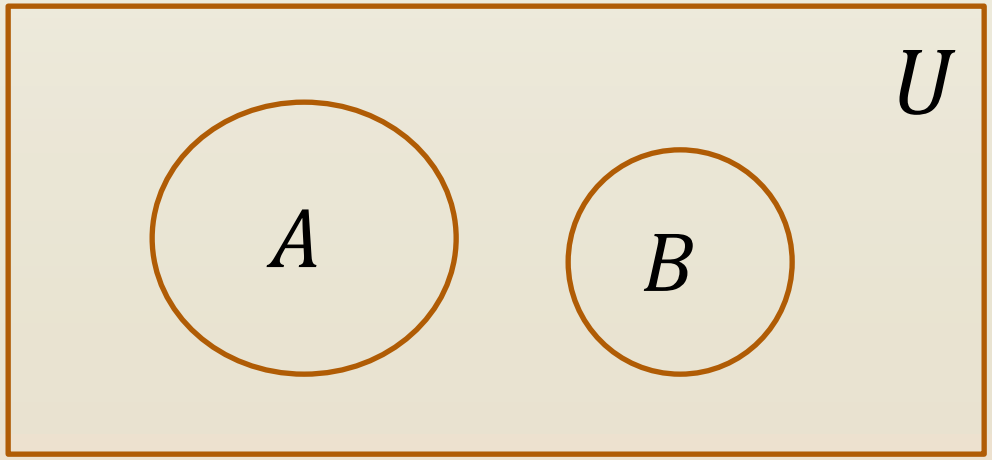
  $C = \{x \mid x \text{ is a positive integer and } x^2 < 12\}$

  then $A = B = C$.

Venn diagrams

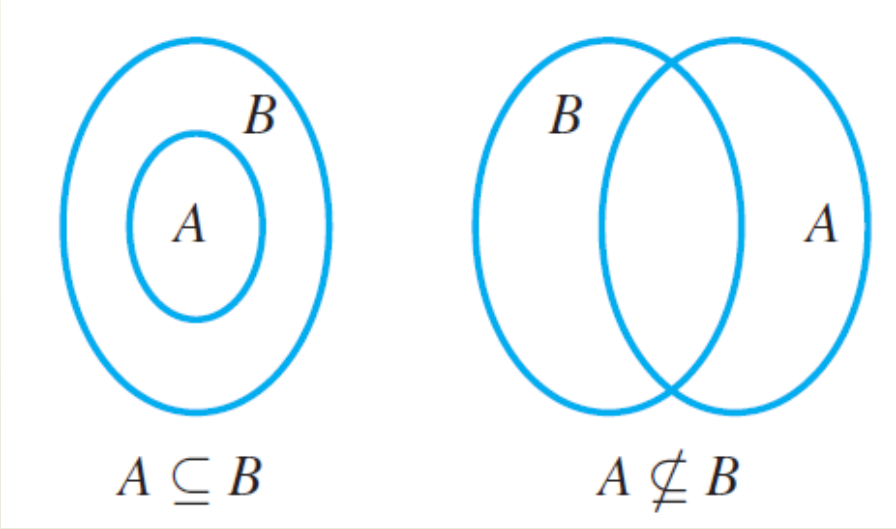
- Universal set U : a set containing all objects for which the discussion is meaningful.

- If we are discussing real numbers and we mention sets A and B, then A and B must be sets of real numbers, not matrices, or other types of objects.



Subsets

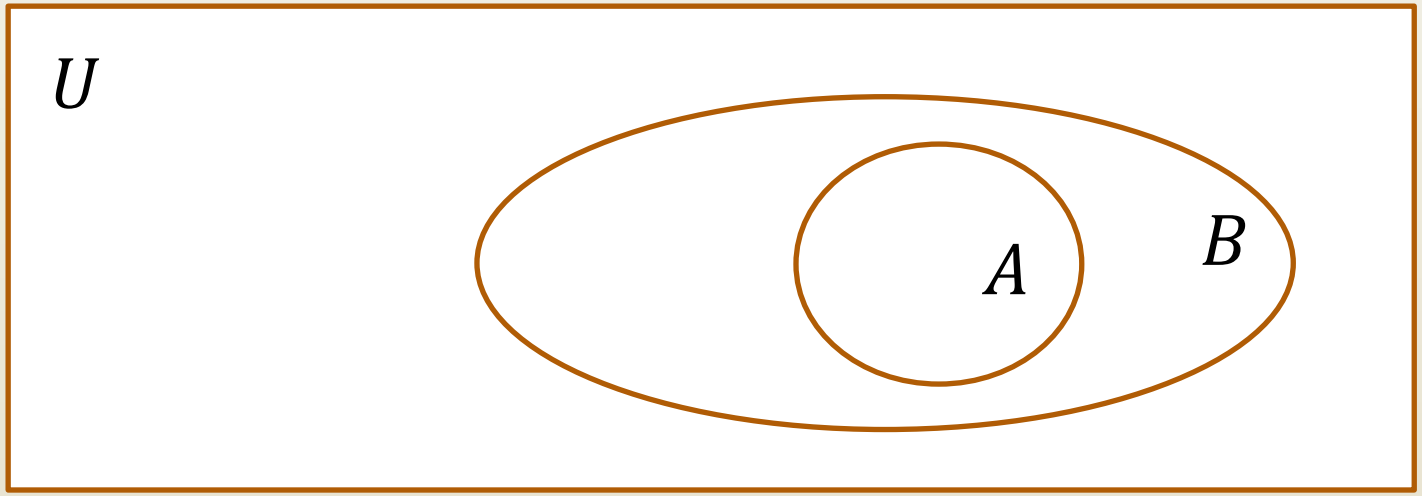
- If every element of A is also an element of B, that is, if whenever, $x \in A$ then $x \in B$, then
  - A is a subset of B
  - A is contained in B
  - $A \subseteq B$



- If A is not a subset of B, then $A \nsubseteq B$
- $\emptyset \subseteq S$ → Empty set is a subset of any set.
- $S \subseteq S$ → Any set S is a subset of itself.

A proper subset

- A set A is said to be a proper subset of B if and only if $A \subseteq B$ and $A \neq B$.
- We denote that A is a proper subset of B with the notation $A \subset B$.



Examples

- $\mathbb{Z}^+ \subseteq \mathbb{Z}$ $\quad\quad$ $\mathbb{Z} \subseteq \mathbb{Q}$
- $A = \{1,2,3,4,5,6\}$ $\;\; B = \{2,4,5\}$ $\;\; C = \{1,2,3,4,5\}$
  - $B \subseteq A$ $\quad\quad$ $B \subseteq C$ $\quad\quad$ $C \subseteq A$
  - $A \nsubseteq B$ $\quad\quad$ $A \nsubseteq C$ $\quad\quad$ $C \nsubseteq B$
- Let A be a set and let $B = \{A, \{A\}\}$
  - $A \in B$ $\quad\quad$ $\{A\} \in B$
  - $\{A\} \subseteq B$ $\quad\quad$ $\{\{A\}\} \subseteq B$
  - $A \nsubseteq B$
- Every set is a subset of itself : $A \subseteq A$
- Empty set is a subset of any set A : $\emptyset \subseteq A$

Finite and infinite sets

- Finite set
  - A set whose elements can be counted.
  - A set has n distinct elements, where $n \in \mathbb{N}$. In this case, $n = |\mathrm{A}|$ = **cardinality** of A
- An infinite set
  - A set whose elements can not be counted.

- Examples of finite sets : $A = \{1,2,3,4,5,6\}$ $B = \{2,4,5\}$ $C = \{1,2,3,4,5\}$
- Examples of infinite sets : $\mathbb{Z}^+, \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$

Cardinality

- The cardinality of S is denoted by $|S|$.
- Ex.
  - V: Vowels in the English alphabet: |V| = ?
  - T: Thai alphabet: |T| = ?
  - X: First seven prime numbers: |X| = ?
  - |∅| = ?

Power set

- $P(A)$: Power set of A → The set of all subsets of A
- Example
  - For a set $\mathrm{A} = \{1\}$, $P(A) = ?$, $|P(A)| = ?$
  - For a set $B = \emptyset$, $P(B) = ?$, $|P(B)| = ?$
  - For a set $C = \{1,2\}$, $P(C) = ?$, $|P(C)| = ?$
  - For a set $D = \{1, \{1\}\}$, $P(D) = ?$, $|P(D)| = ?$
  - For a set $E = \{1,2,3\}$, $P(E) = ?$, $|P(E)| = ?$
  - If F is a set with $|F| = n$, then $|P(F)| = ?$